HDOC DAY-53

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BATCH-15

Question

Write a C program to read the size and elements of a variable-length integer array nums[]. Pass the array to a function product_except_self() using pointers.

The function should calculate the product of all array elements except the current element, without using division, and display the result separated by commas.

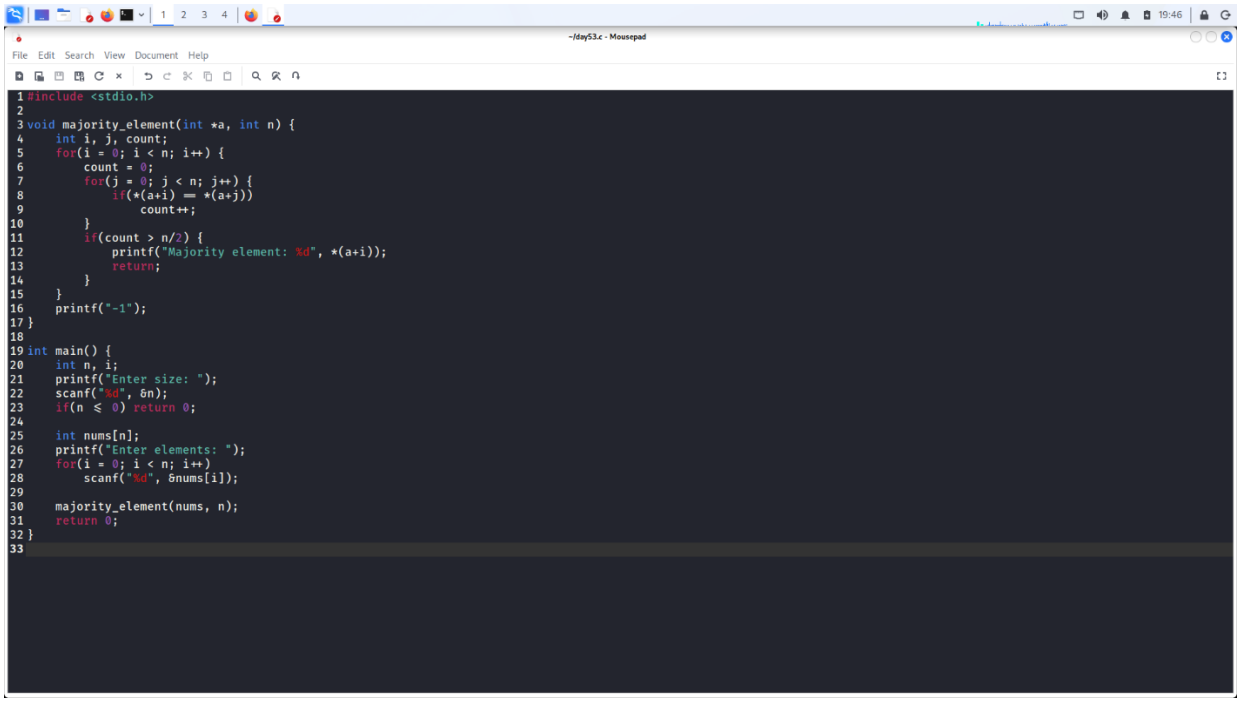
Aim

To write a C program to find the product of all elements of an array except the element at the current index using functions and pointers without division.

Algorithm

1. Start.
2. Read the size n of the array.
3. Read n elements into nums[].
4. Pass the array to product_except_self() using a pointer.
5. Create an output array answer[].
6. Calculate prefix products and store them in answer[].
7. Calculate suffix products and multiply them with answer[].
8. Display the resulting array separated by commas.
9. Stop.

>C PROGRAM:



```c
#include <stdio.h>

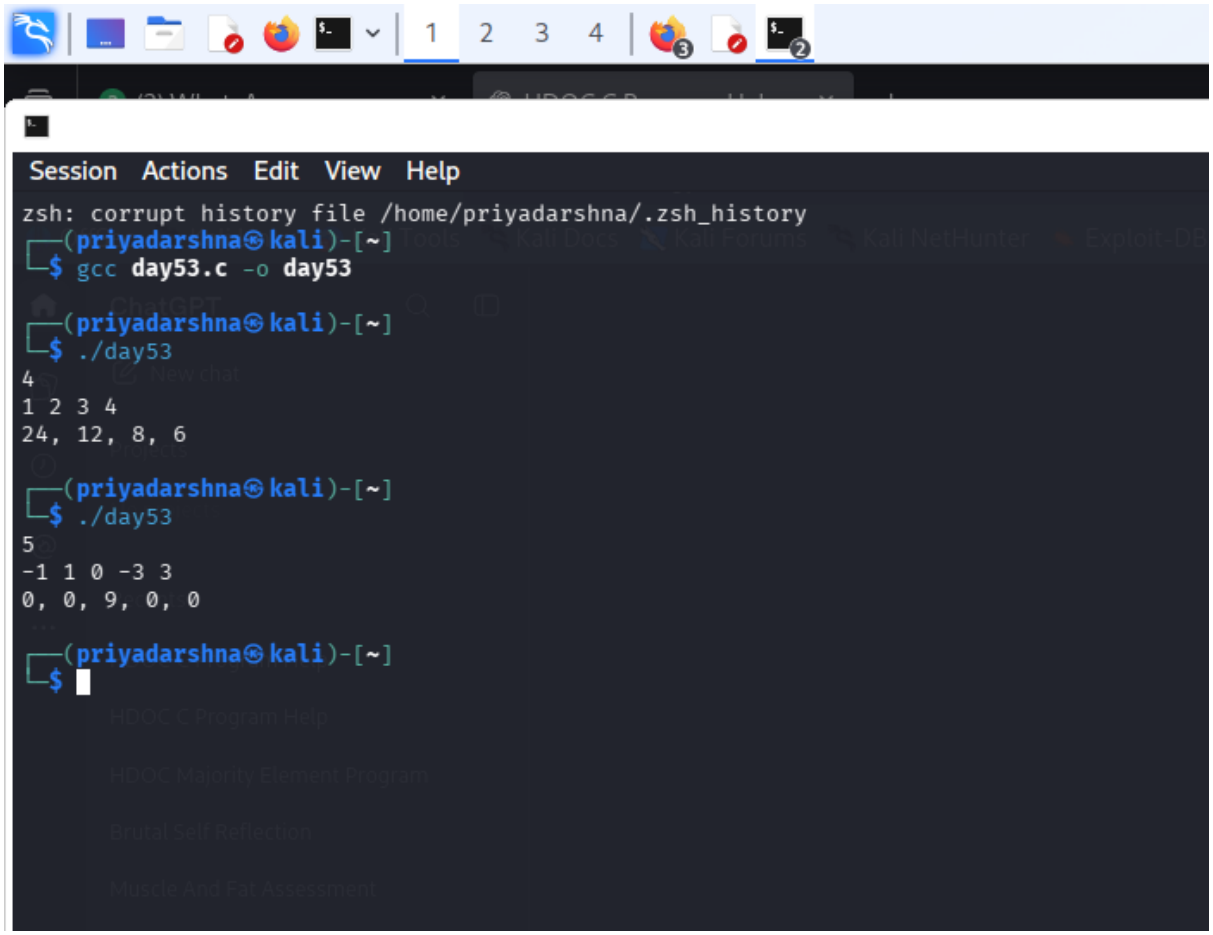
void majority_element(int *a, int n) {
    int i, j, count;
    for(i = 0; i < n; i++) {
        count = 0;
        for(j = 0; j < n; j++) {
            if(*(a+i) == *(a+j))
                count++;
        }
        if(count > n/2) {
            printf("Majority element: %d", *(a+i));
            return;
        }
    }
    printf("-1");
}

int main() {
    int n, i;
    printf("Enter size: ");
    scanf("%d", &n);
    if(n <= 0) return 0;

    int nums[n];
    printf("Enter elements: ");
    for(i = 0; i < n; i++)
        scanf("%d", &nums[i]);

    majority_element(nums, n);
    return 0;
}
```

>OUTPUT:



```
Session  Actions  Edit  View  Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
┌──(priyadarshna㉿kali)-[~]
└─$ gcc day53.c -o day53

┌──(priyadarshna㉿kali)-[~]
└─$ ./day53
4
1 2 3 4
24, 12, 8, 6

┌──(priyadarshna㉿kali)-[~]
└─$ ./day53
5
-1 1 0 -3 3
0, 0, 9, 0, 0

┌──(priyadarshna㉿kali)-[~]
└─$
```